CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EE) Examination
April 2018

EE244 ELECTRICAL POWER GENERATION AND ECONOMICS

Date: 30.04.2018, Monday **Time: 10:00 a.m To 01:00 p.m** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the following questions.** **[07]**

a. Solar thermal power generation can be achieved by______________.

(i) Using focusing collector or heliostates (ii) Using flat plate collectors

(iii) Using solar pond (iv) Any of the above system

b. In fuel cell, the__________energy is converted into electrical energy.

(i) Mechanical (ii) Chemical

(iii) heat (iv) Sound

c. The modern steam turbines are____________.

(i) Impulse turbines (ii) Reaction turbines

(iii) Impulse-reaction turbines (iv) None of the above

d. The draught which a chimney produces is called__________.

(i) Induced draught (ii) Natural draught

(iii) Forced draught (iv) Balanced draught

e. The function of the moderator in a nuclear reactor is to________

(i) Stop chain reaction (ii) Absorb neutrons

(iii) Reduce the speed of neutrons (iv) Reduce temperature

f. A geothermal field may yield_______

(i) Hot water (ii) Wet steam

(iii) Dry steam (iv) All of the above

g. In which of the following power plant the availability of power is least reliable

(i) Tidal power plant (ii) Wind energy

(iii) Solar power plant (iv) Geothermal power plant

Q - 2 (a) Explain the layout of nuclear power plant with suitable diagram. **[06]**

	(b)	With net sketch explain the layout of closed cycle gas turbine power plant.	**[05]**
	(c)	Write down the advantages and disadvantages of thermal power plant.	**[03]**

OR

Q – 2	**(a)**	Explain different types of wind turbine with neat sketch.	**[06]**
	(b)	Write down the advantages & disadvantages of Gas turbine Power Plant.	**[05]**
	(c)	Write down the difference between impulse turbine & reaction turbine.	**[03]**
Q - 3	**(a)**	Draw the schematic diagram of a thermal power station & explain its operation.	**[07]**
	(b)	Explain the layout of hydro power plant in detail with its schematic diagram.	**[05]**
	(c)	Explain the different causes of damage to the boilers and steam turbines with its remedies.	**[02]**

OR

Q - 3	**(a)**	Explain the different types of solar collector used for generation of electricity.	**[07]**
	(b)	Explain the site selection consideration for nuclear power plant.	**[05]**
	(c)	Write down the advantages of Combined Gas Turbine Power Plant.	**[02]**

SECTION – II

Q - 4 **Answer the following questions.** **[07]**

a. Load curve helps in deciding____________.

(i) Total installed capacity of the plant (ii) Sizes of the generating units

(iii) Operating schedule of generating units (iv) All of the above

b. By improving power factor of the system, the kilowatts delivered by the generating station are________.

(i) Decreased (ii) Increased

(iii) Not changed (iv) None of the above

c. Load factor of a power station is generally_____________.

(i) Equal to unity (ii) Less than unity

(iii) More than unity (iv) Equal to zero

d. High tension cables are generally used upto__________.

(i) 11 kV (ii) 33 kV

(iii) 66 kV (iv)132 kV

e. The thickness of the insulation on the conductor, in cables, depends upon________.

(i) Reactive power (ii) Power factor

(iii) Voltage (iv) Current carrying capacity

f. If the shunt capacitance is reduced, then string efficiency is________.

(i) Increased (ii) decreased

(iii) remain same (iv) all of the above

g. In capacitance grading of cables we use a _______dielectric.

(i) Composite (ii) Porous

(iii) Homogeneous (iv) Hygroscopic

Q - 5 (a) Define grading of cable. Explain the any one method of grading for cable. **[06]**

(b) Explain the construction of cable with diagram. **[05]**

(c) Calculate the capacitance and charging current of a single core cable used on a 3-phase, 66 kV system. The cable is 1 km long having a core diameter of 10 cm and an impregnated paper insulation of thickness 7 cm. the relative permittivity of the insulation may be taken as 4 and the supply at 50 Hz. **[03]**

OR

Q – 5 (a) A generating station has the following daily load cycle : **[06]**

Time (hours)	0-6	6-10	10-12	12-16	16-20	20-24
Load (MW)	40	50	60	50	70	40

Draw the load curve and find (i) maximum demand (ii) units generated per day (iii) average load and (iv) load factor.

(b) Define the following terms. **[05]**

i) Connected load ii) Maximum demand iii) Demand factor iv) Average load v) diversity factor.

(c) Why are insulators used with overhead lines? Discuss the desirable properties of insulators. **[03]**

Q - 6 (a) Define depreciation. Explain the any two methods of determining the depreciation of the equipment. **[06]**

(b) List out power factor improvement equipment. Explain any two equipment in detail with necessary diagram. **[05]**

(c) A single phase motor connected to 400 V, 50 Hz supply takes 31.7 A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the the power factor to 0.9 lagging. **[03]**

OR

Q - 6 (a) Define tariff. Explain different types of tariff. **[06]**

(b) A string of 4 insulators has a self-capacitance equal to 10 times the pin to earth capacitance. Find (i) the voltage distribution across various units expressed as a percentage of total voltage across the string and (ii) string efficiency. Refer the given Fig. 1. **[05]**

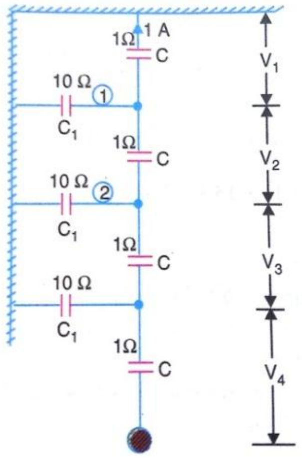

Fig. 1

(c) Explain Suspension type insulators with necessary diagram. **[03]**